Introduction and project motivation

To make a good project of interest I decided to choose World Bank Findex Survey data to use the knowledge on real microdata and make comprehensive analysis and provide economic interpretation.

The Global Findex is the world's most comprehensive database on digital and financial inclusion. I chose my home country as my analysis target.

The code for this project can be found here.

Data

The analysis is based on microdata from the World Bank Global Findex Database (2025 edition). The dataset contains individual-level survey responses related to account ownership, digital payments and demographic characteristics.

The project focuses on Kazakhstan data. It consists of 1000 rows and more than 100 variables but after filtering the data we have in total 1000 rows of data with 44 variables, the most of which are binary.

Project Structure

1. Data cleaning and preprocessing
    a. Removal of empty observations
    b. Transformation of the variable values
    c. Removal of highly correlated attributes
2. Principal component analysis
    a. Identification of variables relevant for clustering
    b. The variable contributions for PC1 and PC2
    c. Visualization of PCA results
3. Classification
    a. Determination of the optimal number of clusters (Elbow and Silhouette methods)
    b. Visualization of clusters using the k-means algorithm
    c. Visualization of clusters using hierarchical clustering
    d. Economic interpretation of the identified groups
4. Demographics
    a. Analysis of demographic differences across clusters
5. Supervised learning
    a. Multinomial Regression
    b. Random Forest

2. Data cleaning and preprocessing

Removal of empty observations. Empty observations and variables with a very high share of missing values were removed. Variables with more than 70% missing value were excluded from the analysis, as they do not provide sufficient information. Responses such as "Don't know" and "Refused" were treated as missing values, as they do not reflect actual financial behavior and cannot be meaningfully interpreted in clustering.

Transformation of the variable values. The raw survey data contain many variables with different formats, missing responses, and special answer categories such as "Don't know" or "Refused". Binary variables were harmonized to represent clear "yes/no" or 0/1 outcomes, while ordinal

variables were converted to ordered numerical scales reflecting increasing or decreasing intensity (for example, frequency of digital payments).

Removal of highly correlated attributes. Finally, variables with very high pairwise correlations (above 70%) were removed to avoid redundancy. This step ensures that highly similar attributes do not disproportionately influence the analysis and allows each remaining variable to contribute unique information.

This way to avoid redundancy and multicollinearity in the dataset, the following variables were removed:

Original variable(s)	Kept variable
Made or received a digital payment (anydigpayment)	Has an account (account)
Has a digitally enabled account (dig_account)	
Made a digital merchant payment (merchantpay_dig)	
Used a card or mobile phone to access account (fin3)	
Used a mobile phone or a card to pay for an in-store purchase (fin25e2)	
Has an account at a financial institution (account_fin)	
Fin32 Received a wage payment	receive_wages
Received financial support from the government Fin37	receive_transfers
Fin38 Received a public sector pension	receive_pensions
Fin42 Received an agricultural payment	receive_agriculture
Fin30 Made a utility payment	pay_utilities
Fh1 Made a domestic remittance	domestic_remittances
Fh2 Received a domestic remittance	

3. Principal Component Analysis

Identification of variables relevant for clustering. Demographic and nominal variables were not included in the analysis to obtain results based only on behavioral patterns and not demographic attributes and age, education etc.

After preprocessing, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset and to identify the main underlying patterns in financial behavior.

The variable contributions for PC1 and PC2.

The top 10 variables for Principal Component 1

Variable	Definition	PC1 contribution (%)
fin20	Applied for a loan using a mobile phone (yes/no)	30,7
borrowed	The respondent, personally or together with someone else, borrowed money in the past year, including from a bank or similar financial institution, via a mobile money account, from family or friends, or from an informal savings group, or for any other reason (yes/no)	30,3
fin22a	Borrowed from a formal bank or similar financial institution (yes/no)	30,4
fin10	Has a credit card (yes/no)	28,3
receive_wages	Received a wage payment (yes/no)	25,3
account_mob	Has a mobile money account (yes/no)	24,5
fin22b	Borrowed from friends or family (yes/no)	22,5
internet_use	Using the internet within the past three months (yes/no)	21,1

domestic_remittances	Made or received a domestic remittance payment (yes/no)	20,9
fin19	Made regular payments to insurance agent or company	20,4

The top 10 variables for Principal Component 2

Variable	**Definition**	**PC2 contribution (%)**
fin6	Frequency of money deposited into accounts	-37,0
account	Has an account (yes/no)	-35,7
fin25e1	Use a mobile phone or card to pay for household cleaning supplies (yes/no)	32,8
fin26b	Used a mobile phone or the internet to buy something online (yes/no)	32,2
fin8	Store money in account (yes/no)	31,0
fin2	Has a debit card (yes/no)	-29,0
fin26a	Used a mobile phone or computer to make a bill payment	28,2
fin9b	Check account balance using a mobile phone or the internet (yes/no)	27,9
fin25e3	Frequency of digital instore merchant payments (frequency)	-20,1
fin4	Opened first account to receive a wage or salary payment or money from the government (yes/no)	15,2

PC1 = "Financial engagement and borrowing behavior"

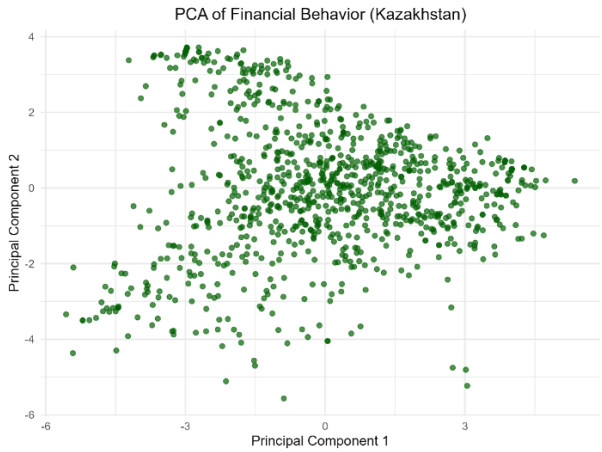
Individuals with high PC1 scores demonstrate active financial engagement and intensive borrowing behavior. They are more likely to:

1. use loan products, including formal and digital loans
2. borrow money from financial institutions or via mobile banking and other
3. own credit cards and mobile accounts
4. receive wages and make domestic remittance or insurance payments

PC2 = "Digital vs traditional financial behavior"

Individuals with high PC2 scores are characterized by active digital transaction usage rather than passive account ownership. They are more likely to:

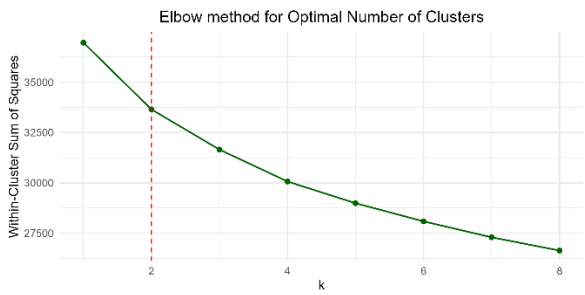
1. actively use digital payment methods and check balance online
2. purchase products online
3. use cards and mobile-based accounts for transactions
4. participate in receiving pension payments through digital or account-based channels



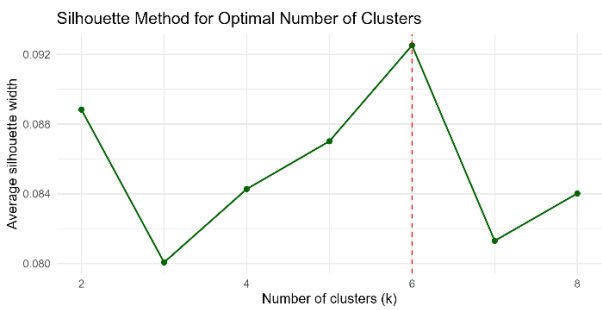
4. Classification

Determination of the optimal number of clusters (Elbow and Silhouette methods).

The Elbow method indicates that the optimal number of clusters is k = 2. However, this solution leads to a very trivial interpretation, separating individuals into *financially included* and *financially excluded* groups. As this binary distinction does not provide sufficient insight, the two-cluster solution is not used for further analysis.



The Silhouette method was also applied to assess cluster quality. The highest average silhouette widths were observed for k = 2 and k = 6, suggesting that both solutions provide relatively well-separated clusters.



Despite these results, the analysis focuses on k = 3 and k = 4 clusters. These configurations offer a better balance between statistical validity and interpretability, allowing for a more meaningful economic interpretation

Visualization of clusters using the k-means and hierarchical clustering algorithms

K-means **Hierarchical clustering**

	Hierarchical clustering		
k-means	150	231	3
	8	444	2
	3	42	120

	Hierarchical clustering			
k-means	382	0	18	42
	4	120	0	0
	123	2	149	6
	40	3	1	110

Economic interpretation of the identified groups for k-means algorithm

	Cluster1	Cluster2	Cluster3
Frequency of money sent or taken out of accounts (fin6) 1 – weekly 2 – monthly 3 – less than once a month 4 – never	1,86	1,42	1,01
Has a debit card (fin2)	85,9%	88,8%	7,4%
Has an account	100%	100%	34%
Used a mobile phone or the internet to buy something online (fin26b)	35,9%	77,3%	98,7%
Borrowed	35,9%	96,4%	43%
Applied for a loan using a mobile phone (fin20)	6,7%	65,2%	6,1%
Has a mobile money account	38,5%	72,6%	15,4%

Borrowed from a formal bank or similar financial institution	5,4%	58,1%	3%
Has a credit card	9,6%	54,1%	0%
Received a wage payment	45,1%	84,6%	34,0%

Cluster 1 (red) — "Basic account holders with limited digital engagement"

This cluster is characterized by high ownership of basic financial products, such as bank accounts and debit cards, but relatively low engagement with digital and advanced financial services.

Main characteristics:

1. All individuals have a bank account (100%) and most have a debit card (85%).
2. However, usage of digital financial services is limited: only 34% used the internet or a mobile phone for online purchases, and very few applied for loans via mobile phones (7%).
3. Borrowing behavior is moderate (44%), but borrowing from formal financial institutions is rare (4%).
4. Credit card ownership is almost nonexistent (1%).
5. Wage receipt is relatively low (34%).

This cluster represents individuals who are financially included at a basic level, but remain largely traditional users of financial services, with limited adoption of digital tools and formal credit.

Cluster 2 (blue) — "Digitally and financially active users"

The second shows consistently high values across almost all indicators, suggesting a group that is both financially sophisticated and digitally engaged.

Main characteristics:

1. All individuals have bank accounts (100%), and most have debit cards (89%).
2. Digital activity is very high: 78% made online purchases and 64% applied for loans using mobile phones.
3. Borrowing is intense (96%), with a substantial share borrowing from formal institutions (57%).
4. Ownership of mobile money accounts (72%) and credit cards (53%) is common.
5. This cluster has the highest share of individuals receiving wages (84%).
6. Money is moved through accounts relatively frequently (mean = 1.49, where 1 is frequent).

This cluster can be described as highly financially included and digitally advanced, representing individuals who actively use a wide range of financial products, including digital payments and formal credit.

Cluster 3 (green) — "Partially included and irregular users"

The third cluster displays mixed characteristics, with lower ownership of traditional banking products but moderate engagement with some financial services.

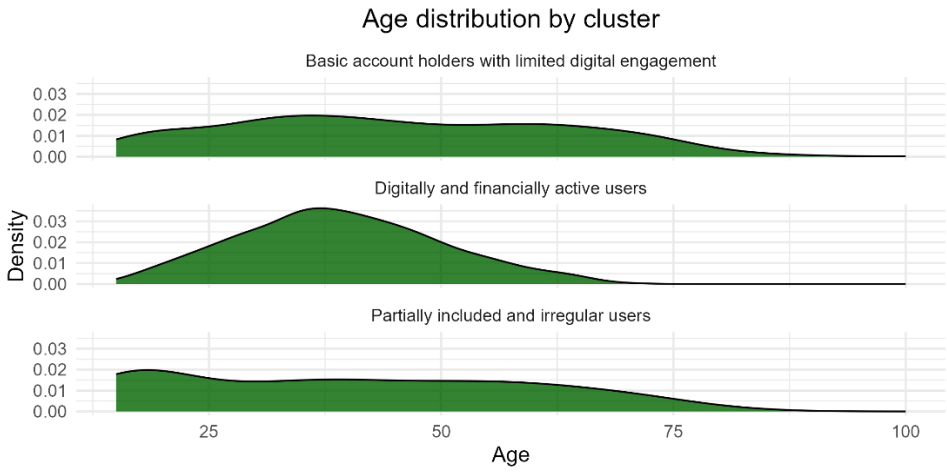
Main characteristics:

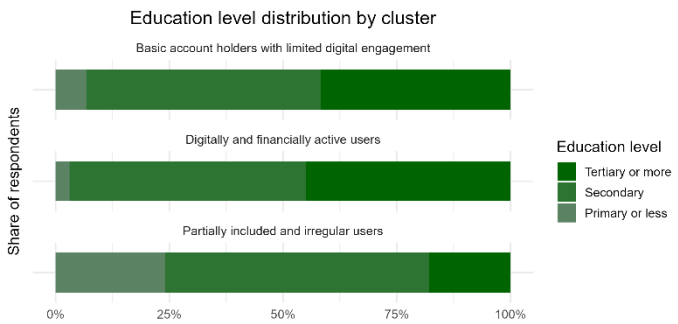
1. Only 35% have a bank account and very few own debit or credit cards.
2. Digital purchasing is limited (34%), like in the first cluster.
3. Borrowing exists (35%), and notably, borrowing from formal institutions (53%) is higher than in Cluster 1.
4. Mobile money account ownership is relatively moderate (38%).

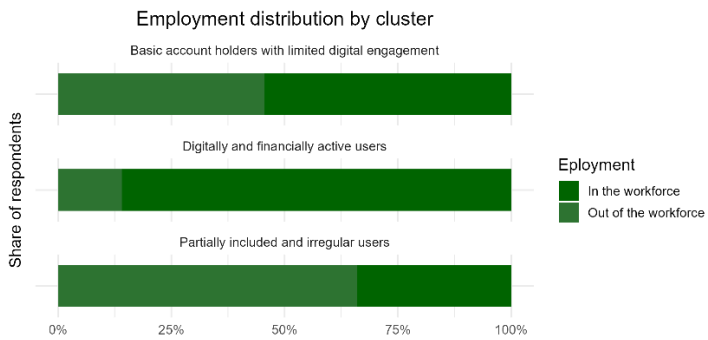
5. Wage receipt is moderate (45%).
6. Frequency of account use is lower than in the second cluster (mean = 1.90).

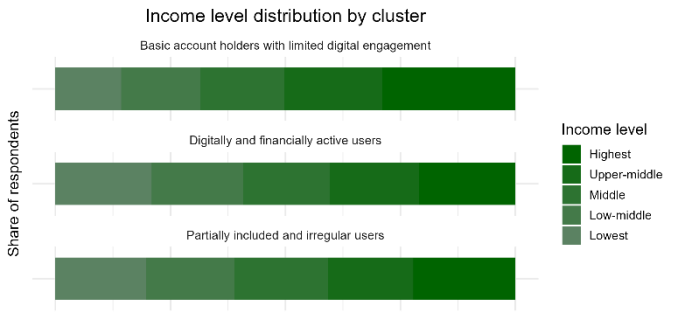
This cluster represents individuals who are partially financially included, possibly relying on specific financial services (such as formal loans or mobile money) without broad access to traditional banking infrastructure.

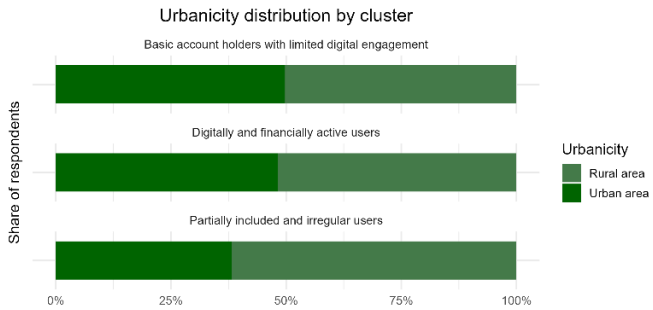
4.Demographics











Cluster 1 "Basic account holders, limited engagement")

- older on average (45.4 years),
- with a slight majority of women (60%)
- moderate employment (54%) and
- a balanced urban/rural composition
- shows a higher concentration in the top income quintile (29%)
- a relatively high share of tertiary education (42–45%) and low primary education

Cluster 2 ("Digitally and financially active users")

- is younger (39.2 years) with a big spike of 30–40-year-old people
- highly employed (86%)
- a balanced urban/rural composition
- a uniform distribution across all income quintiles
- a relatively high share of tertiary education (42–45%) and low primary education

Cluster 3 ("Partially included and irregular users") has

- lower employment (34%)
- a higher share of women (65%)
- a more rural composition, indicating irregular financial activity
- higher proportion of individuals with only primary education (24%) a smaller share with tertiary education (18%) corresponding to lower inclusion and irregular usage
- slightly concentrated in the lower- and middle-income quintiles

The three clusters reveal distinct financial behavior profiles:
Cluster 1 represents older individuals with limited engagement but higher income.
Cluster 2 includes young employed digitally and financially active users.
Cluster 3 captures partially included irregular users with lower employment, education, and income, mostly in rural areas.

Supervised learning

In this part of the analysis, we focus on supervised machine learning methods to predict the sources from which individuals in Kazakhstan obtain emergency funds. Several models were used to compare different results and their consistency.

Supervised models are trained using cross-validation and splitting methods to ensure robust performance estimates, and predictions are evaluated using metrics such as accuracy, sensitivity, specificity, and balanced accuracy. Random Forest and Multinomial Logistic Regression have been used to explore classification performance and variable importance.

Target. The target variable is the main source of emergency money in 30 days.

The full data with all variables was used to compare it with the only-essential-variables approach.

Multinominal regression

The results were obtained by using all the data and splitting on the training and testing data with 70/30 ratio and different number of classes of the target variable.

Class	Sensitivity	Specificity	Balanced Accuracy
Savings	0.3571429	0.9218750	0.6395089
FamilyRelativesFriends	0.6146789	0.6507937	0.6327363
Other	0.3421053	0.7882883	0.5651968
Borrowing	0.4146341	0.9416342	0.6781342
CouldNotComeUpWithmoney	0.3000000	0.9402985	0.6201493

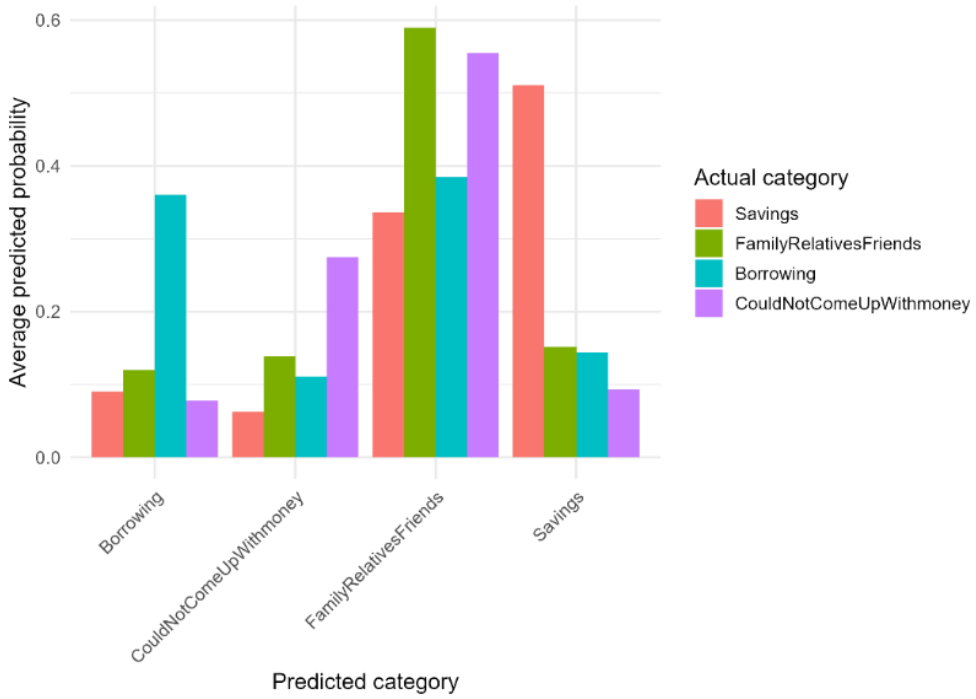
Class	Sensitivity	Specificity	Balanced Accuracy
Savings	0.4047619	0.8862745	0.6455182
FamilyRelativesFriends	0.5137615	0.6010638	0.5574126
MoneyFromWorking	0.2222222	0.8806584	0.5514403
Borrowing	0.4390244	0.8789062	0.6589653
Other	0.0952381	0.9673913	0.5313147
CouldNotComeUpWithmoney	0.1666667	0.9475655	0.5571161

These are results of the CV method with 5 folds and removing the “Other” category.

Class	Savings	Family Relatives Friends	Borrowing	Could’t come up with money
Sensitivity	0.36170	0.6181	0.46377	0.33981
Specificity	0.89421	0.5445	0.88651	0.90047
Balanced Accuracy	0.62796	0.5813	0.67514	0.62014

Class	Savings	Family Relatives Friends	Money From Working	Borrowing	Could’t come up with money
Sensitivity	0.25532	0.5769	0.25275	0.47826	0.23301
Specificity	0.92757	0.5975	0.87131	0.87342	0.92000
Balanced Accuracy	0.59145	0.5872	0.56203	0.67584	0.57650

Visualization of average predicted probability.



As we can see across all specifications, the multinomial logistic regression models demonstrate moderate predictive performance, with balanced accuracy values typically ranging between 0.55 and 0.68, depending on the number of target classes, validation strategy, and variable selection.

Comparing models using all available predictors versus selected variables reveals only modest differences in performance:

- Selected-variable models retain most of the predictive power observed in full models
- In some cases, variable selection improves interpretability without materially reducing balanced accuracy
- This suggests that a smaller subset of well-chosen financial and demographic indicators captures most of the predictive signals

Across all models, a consistent pattern emerges:

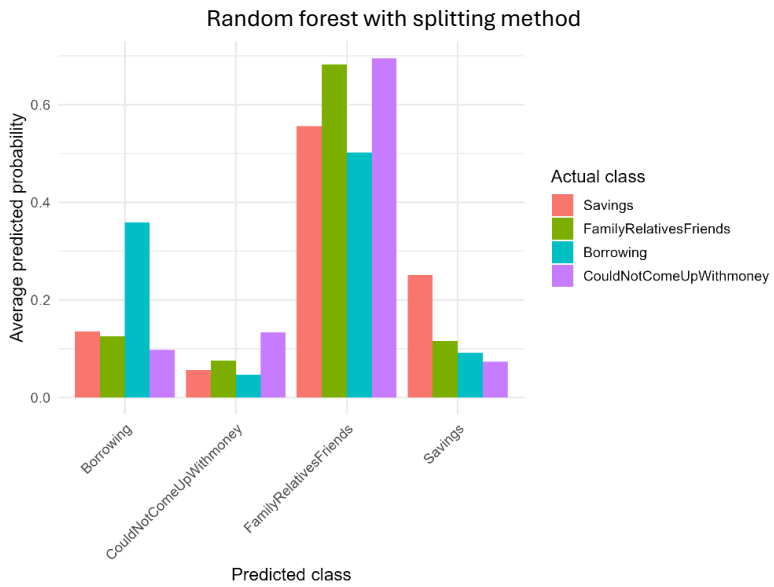
- Specificity is high (often above 0.87), indicating strong ability to correctly identify non-members of a given class

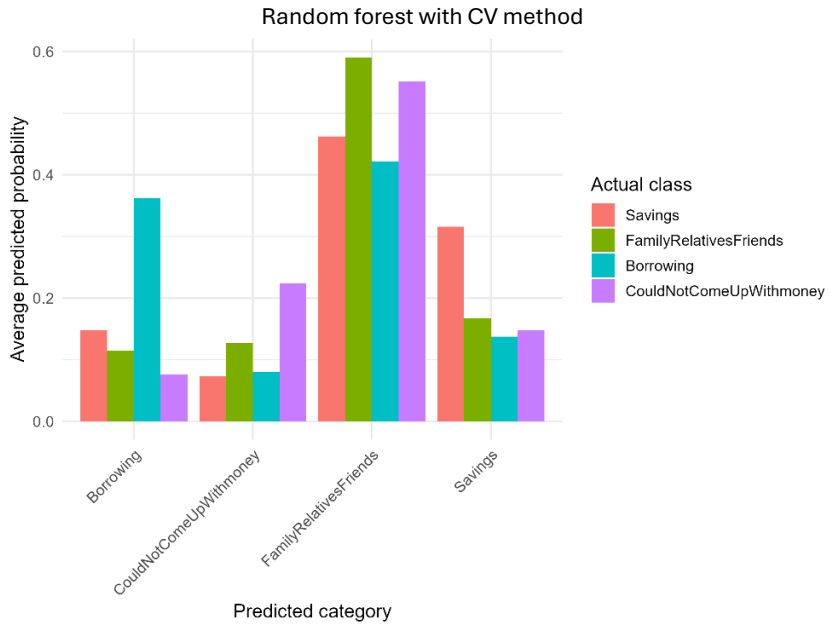
- Sensitivity is comparatively lower, especially for smaller or less distinct classes

Random forest

Class	Sensitivity	Specificity	Balanced Accuracy
Savings	0.35714	0.90556	0.63135
FamilyRelativesFriends	0.81650	0.41590	0.61620
Borrowing	0.39024	0.93923	0.66474
CouldNotComeUpWithmoney	0.10000	0.97396	0.53698

Random Forest variable importance indicates that core financial behaviors (savings, borrowing from formal institutions, pensions, and other transaction patterns) are the strongest predictors of the main source of emergency funds. Demographic variables like age, education, and employment contribute moderately. Predictive patterns differ by class: for example, savings and pensions are key for predicting the 'Savings' category, while formal borrowing strongly predicts 'Borrowing'.





Among the target classes, “Family/Relatives/Friends” demonstrates the highest specificity across all models, indicating that the algorithm is particularly effective at identifying individuals who do not rely on this source of emergency funding. Those not classified under this group are more likely to utilize alternative sources such as borrowing or personal savings, reflecting the model’s ability to distinguish between different financial coping strategies.

Variable	**Most predictive for class(es)**	**Notes / Interpretation**
Difficulty of emergency funds in 30 days	Savings, Borrowing, CouldNotComeUpWithMoney	Highly influential across multiple classes; indicates financial knowledge strongly affects emergency funding behavior.
Received a government pension payment	Savings, Borrowing	Positive impact on Savings; indicates pension recipients are more likely to rely on their savings.
Borrowed from a formal bank or similar financial institution	Borrowing, FamilyRelativesFriends	Strong predictor, especially for Borrowing; reflects engagement with financial systems.
Education level	CouldNotComeUpWithMoney, FamilyRelativesFriends	Higher education helps predict access to or avoidance of emergency borrowing.
Income quartile	Savings, CouldNotComeUpWithMoney	Higher income predicts Savings use; lower income predicts inability to cover emergencies.
Received a wage payment	Borrowing	Those receiving wages sometimes rely on borrowing, likely due to cash flow timing.
Made or received a domestic remittance payment	FamilyRelativesFriends	Receiving remittances increases likelihood of relying on family/friends.
Respondent is in workforce	Savings, Borrowing	Employment status affects ability to rely on savings vs borrowing.
Borrowed	Borrowing	Past borrowing behavior strongly predicts borrowing in emergencies.
Female	Savings, Borrowing	Gender has moderate influence; women more likely to use certain sources.
Made regular payments to insurance agent or company Saved using a savings club or a person outside the family	Multiple classes	Moderate influence; represent other financial literacy/behavior measures.